

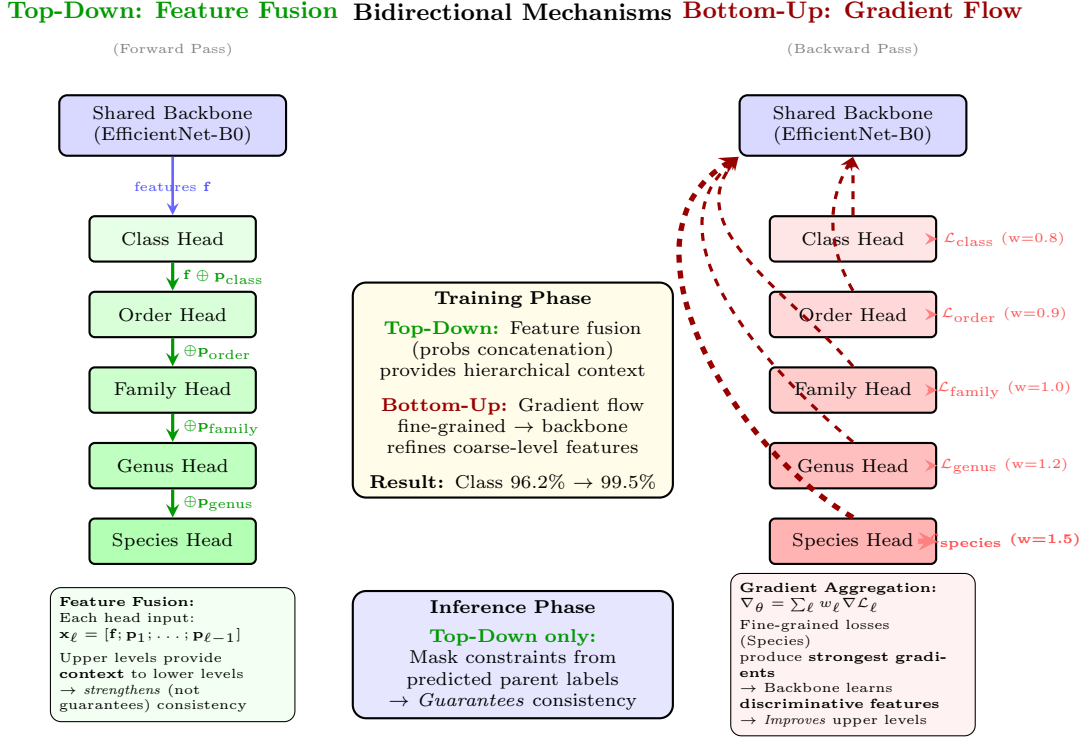


Figure 1: Bidirectional information flow in hierarchical classification. **Left (Top-Down):** During forward pass, feature fusion concatenates backbone features with parent-level probability distributions, providing hierarchical context to each head. This *strengthens* (but does not guarantee) consistency during training. **Right (Bottom-Up):** During backward pass, gradients from all losses converge to the shared backbone. Fine-grained levels (Species, Genus) have higher loss weights, producing stronger gradients that refine backbone features, indirectly improving coarse-level (Class, Order) accuracy. **Center:** Summary of bidirectional mechanisms during training and inference phases.